

# Module 07 — Safer & smarter

Claude Code 101 · Beginner Workshop · Module 7 of 8

The model is helpful, polite, and remembers what you paste. Three things you should always think twice before sending.

## What you'll learn

By the end of this 25-minute lesson you will be able to:

1. List the three categories never to paste into Claude: secrets, PII, and proprietary data you don't own.
2. Recognise them in a real file before you copy.
3. Choose the right substitute: a fake value, a redacted snippet, or a "don't share" decision.

## Why this matters

- Anything you paste leaves your machine. Even if the provider promises not to train on it, "we promise" is not a security control.
- Secrets in chat logs end up in screenshots, in backups, in support tickets. The blast radius is much larger than the chat window.
- Most accidents are not malicious — they happen when someone copy-pastes a config file at 4:55pm. A 10-second redaction habit prevents 100% of those accidents.

# The one concept

Three categories. Three reactions. No exceptions.

| Category                       | Examples                                                                  | What to do instead                                                      |
|--------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Secret</b>                  | API key, password, token, private SSH key, database URL with creds        | Replace with <code>EXAMPLE_KEY</code> or <code>xxx</code> . Then paste. |
| <b>PII</b>                     | Real name + email, phone, address, customer record, internal user list    | Replace with <code>Alice / alice@example.com</code> . Then paste.       |
| <b>Proprietary / under NDA</b> | Customer source code you don't own, internal designs, unannounced product | Don't paste. Describe the shape ("a 50-line auth handler in Go").       |

## Show me

**Wrong** — a real connection string with a real password:

```
> Why does this connection fail?  
DATABASE_URL=postgres://admin:Hunter2-real-password@prod-db.acme.com:5432/customers
```

**Right** — same shape, no actual secrets:

```
> Why does this connection fail?  
DATABASE_URL=postgres://USER:PASSWORD@HOST:5432/DBNAME
```

Claude can debug the second one just as well as the first. The first one might end up in a support transcript, a screenshot, or a colleague's chat history.

## Try it yourself

`exercises/beginner/part-07/starter/leaky_config.txt` contains a fake config file with three planted leaks. Your job: redact all three and save the cleaned version. The exercise asks Claude **only** about the redacted version.

Time budget: 10 minutes. None of the leaks are real — they are obvious placeholders so nobody can be hurt if you slip.

## Common mistakes

- **"It's just localhost"** — `localhost` URLs can still contain real tokens in query strings.
- **"It's just a test account"** — test accounts share infrastructure with prod more often than you think.
- **"The model promised not to train on it"** — irrelevant. The risk is logs, screenshots, and pastes-of-the-paste.
- **Redacting only the obvious one** — config files leak in surprising places (comments, env vars, file paths).
- **Asking Claude to redact it for you, then sending the original to be redacted.** That's the leak.

# Lesson reflection

Take 90 seconds:

1. In the last week, did you paste anything into a chatbot that fit one of the three categories?
2. Look at your last terminal scrollback. Is there anything you'd be uncomfortable sharing on Slack? That's the same standard.
3. If a colleague asked you "is it safe to paste this?", what one-sentence rule would you give them?

## What's next

Module 08 — **Putting it together** — is the capstone. You'll build a tiny Python CLI using everything from Modules 01–07, and a script will grade your result.

Budget for Module 08: 30 minutes.

## Glossary card

- **Permission:** A yes/no setting that lets a tool read, write, or run things on your machine.
- **PII:** Personally identifiable information — names, addresses, phone numbers, account numbers, and similar private data.
- **Secret:** Sensitive data like a password or API key that must never be pasted into a prompt.